

Selection of appropriate statistical tests

Dr Sandheep Sugathan
MD DPH FAIMER Fellow
Senior Lecturer in Public Health
UniKL RCMP

Dependent variable	Independent variable	Test of statistical significance
Categorical variable Diabetes status (Diabetic or non diabetic)	Categorical variable Obesity (Obese / overweight / normal / underweight)	Chi square test

Dependent variable	Independent variable	Test of statistical significance
Categorical variable Anemic status (Anemic / normal)	Categorical variable Household income (low / medium / high)	Chi square test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is normally distributed	Categorical variable with 2 independent categories – (gender / obesity status / diabetes status / hypertensive status) comparing means between 2 independent groups	Independent samples T test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is normally distributed	Categorical variable with 2 related categories – (before and after treatment / before and after experiment – in the same group of patients) comparing means between 2 related groups (pre and post)	Paired samples T test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is normally distributed	Categorical variable with more than 2 categories – (ethnicity / education status / occupation groups) comparing means between more than 2 groups	ANOVA test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is normally distributed	Comparing with a standard value comparing mean of a sample with a standard value (obtained from a text book / clinical practice guideline)	One sample T test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is not normally distributed	Categorical variable with more than 2 categories – (ethnicity / education status / occupation groups) comparing the dependent continuous variable between more than 2 groups	Non parametric test / Kruskal wallis test

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is not normally distributed	Categorical variable with 2 independent categories – (gender / obesity status / diabetes status / hypertensive status) comparing means between 2 independent groups when data is skewed	Non parametric test (Wilcoxon Signed Rank test / Mann Whitney U test)

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is not normally distributed	Categorical variable with 2 related categories – (before and after treatment / before and after experiment – in the same group of patients) comparing means between 2 related groups (pre and post) when data is skewed	Non parametric test (Wilcoxon Signed Rank test / Mann Whitney U test)

Dependent variable	Independent variable	Test of statistical significance
Continuous variable -Height / Weight / BMI / GFR / Serum cholesterol / Fasting Blood Glucose Data is not normally distributed	Categorical variable with more than 2 independent categories – (ethnicity / education status / occupation groups) comparing means between more than 2 independent groups when data is skewed	Non parametric test (Kruskal Wallis test)